1. Key Injection Analysis

The `FF 82` APDU for key injection used in your script:

```plaintext
FF 82 00 00 10 [Derived Key Bytes]
```

Matches the `Load Authentication Keys` command described in the ACR122U documentation (page 12):

- P1: `00` - Load the key into volatile memory.
- P2: `00` - Key location.
- Lc: `06` for MIFARE keys or `10` for other keys.

However, this APDU is specific to MIFARE Classic cards, not EMV cards or ISO 14443-4 tags.

2. APDU for EMV-Compatible Cards

For EMV-compatible or ISO 14443-4 cards, follow the ISO 7816-4 APDU standards:

1. **Select Application**:
   ```plaintext
   00 A4 04 00 Lc [AID]
   ```

2. **Retrieve TLV Data**:
   ```plaintext
   80 CA 9F 17 00
   ```

Your `select_application` function aligns well with this standard:

```python
apdu = [0x00, 0xA4, 0x04, 0x00, len(aid_bytes)] + list(aid_bytes) + [0x00]
```

3. Validate Card Type Before Key Injection

Before using `FF 82`, confirm the card type with the Get Data APDU (`FF CA`).

Updated Function for Card Type Detection:

```python
def get_card_type(connection):
    apdu = [0xFF, 0xCA, 0x01, 0x00, 0x00]  # Get ATS
    try:
        response, sw1, sw2 = connection.transmit(apdu)
        if sw1 == 0x90 and sw2 == 0x00:
            print(f"Card Type Identified: {toHexString(response)}")
            return response
        else:
            print(f"Failed to retrieve card type: SW1={sw1:02X}, SW2={sw2:02X}")
    except Exception as e:
        print(f"Error retrieving card type: {e}")
    return None
```

If the card is MIFARE Classic:

- Use `FF 82` to load keys.
- Use `FF 86` for block authentication.

If the card is EMV/ISO 14443-4:

- Use ISO 7816-4 APDUs such as `00 A4` and `80 CA`.

4. Direct Transmit for Non-Standard Tags

If the reader rejects the `FF 82` APDU, use Direct Transmit to send raw payloads:

```plaintext
FF 00 00 00 [Lc] [Payload]
```

Example:

```python
apdu = [0xFF, 0x00, 0x00, 0x00, len(payload)] + payload
response, sw1, sw2 = connection.transmit(apdu)
```

5. Updated Workflow

1. Detect Card Type:
   - Use `FF CA` to get the UID or ATS.

2. For MIFARE Classic Cards:
   - Load keys with `FF 82`.
   - Authenticate with `FF 86`.

3. For EMV-Compatible Cards:
   - Use standard ISO 7816-4 APDUs:
     - SELECT (`00 A4`).
     - GET DATA (`80 CA`).

4. Direct Transmit:
   - Send custom APDUs for unsupported operations.

By implementing card type detection and aligning APDUs with the correct standard, the code will be compatible with both MIFARE and EMV cards.